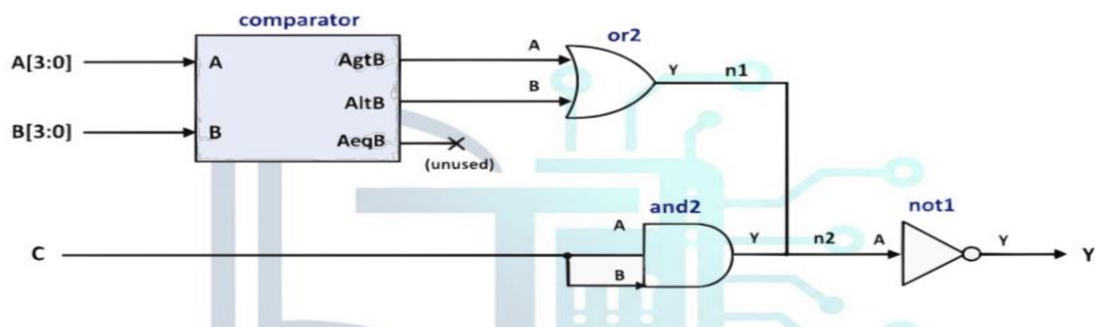
VERSION	ACTION	DATE	AUTHOR
1.0	Initial draft	05/07/2026	Abdulmalek Aldossery

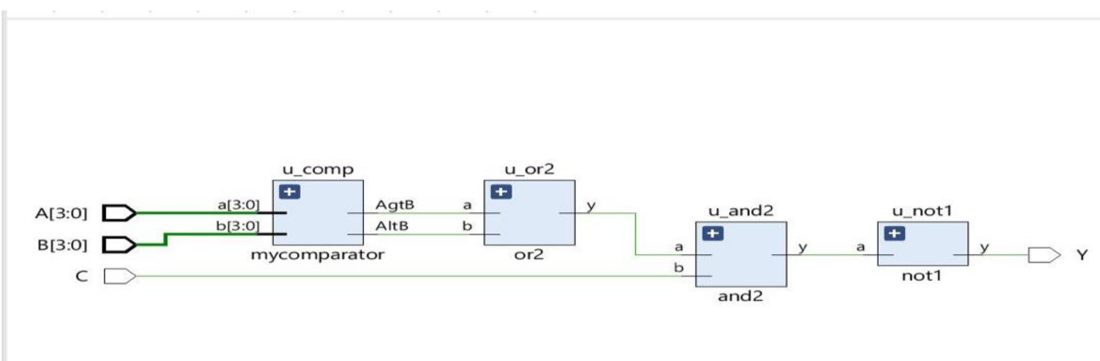
Task #1.1

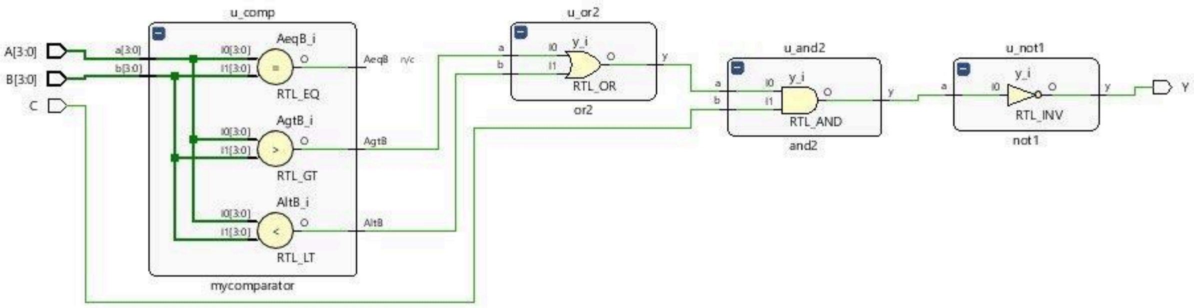
Using SystemVerilog, implement the following design by connected the blocks <u>(USE the design source below)</u>



comparator.sv:

```
module comparator(AeqB, AltB, AgtB, a, b);

input [3:0] a,b;
output AeqB, AltB, AgtB;

assign AeqB = (A == B)? 1 : 0;
assign AltB = (A < B)? 1 : 0;
assign AgtB = (A > B)? 1 : 0;
endmodule
```





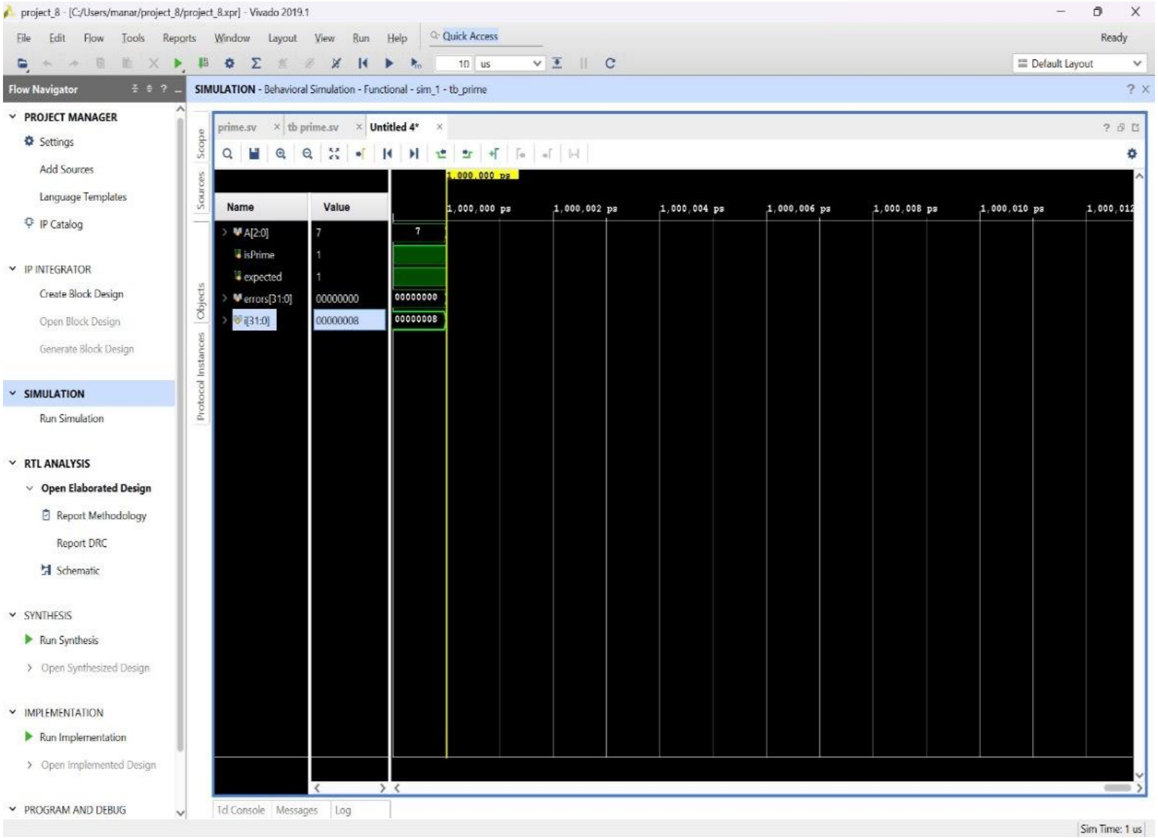
Task #1.2

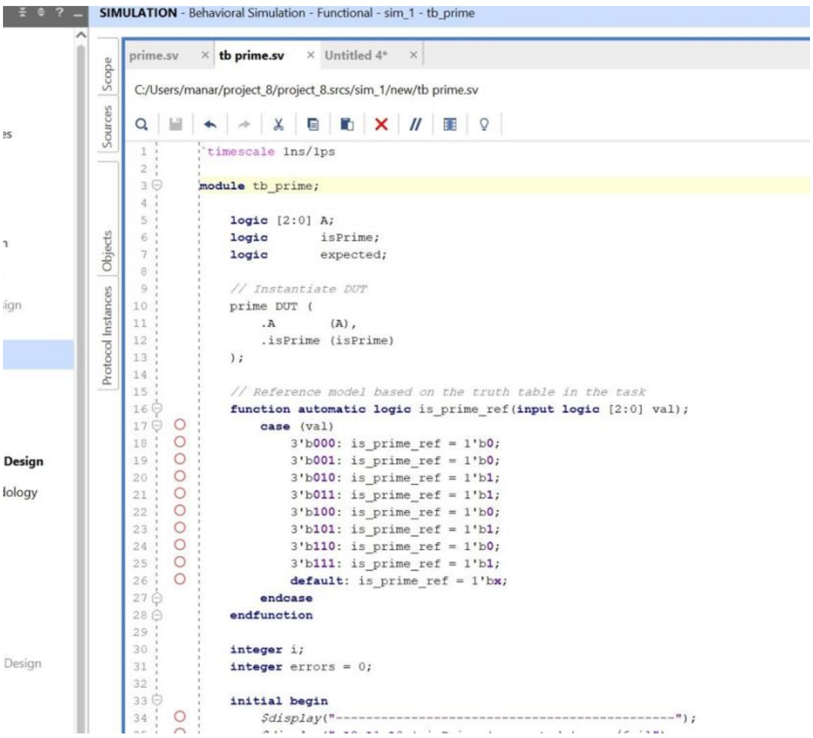
Given the following SystemVerilog design for prime Circuit that check the 3-bit binary number is prime or not (Prime numbers are those number that can be divided by only 1 or itself like 2,3,5,7). <u>You are asked to write the test bench to verify the design (USE the design source below)</u>:

A2	*A1*	*A0*	*isPrime*
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

prime.sv:

```
module prime(

input logic [2:0] A,
 output logic isPrime

);

assign isPrime = (A[2])? A[0]: A[1];

endmodule
```

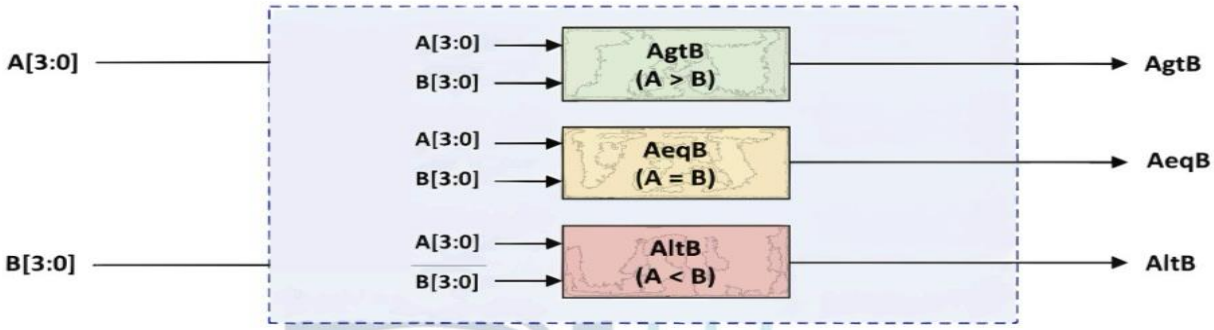

project_8 - [C:/Users/manar/project_8/project_8.xpr] - Vivado 2019.1
File Edit Flow Tools Reports Window Layout View Run Help
Quick Access
Ready
Default Layout
Flow Navigator
SIMULATION - Behavioral Simulation - Functional - sim_1 - tb_prime
PROJECT MANAGER
Settings
Add Sources
Language Templates
IP Catalog
IP INTEGRATOR
Create Block Design
Open Block Design
Generate Block Design
SIMULATION
Run Simulation
RTL ANALYSIS
Open Elaborated Design
Report Methodology
Report DRC
Schematic
SYNTHESIS
Run Synthesis
Open Synthesized Design
IMPLEMENTATION
Run Implementation
Open Implemented Design
PROGRAM AND DEBUG
prime.sv
tb prime.sv
Untitled 4*
Name
Value
A[2:0] 7
isPrime 1
expected 1
errors[31:0] 00000000
i[31:0] 00000008
1,000,000 ps
1,000,002 ps
1,000,004 ps
1,000,006 ps
1,000,008 ps
1,000,010 ps
Tcl Console
Messages
Log
Sim Time: 1 us


SIMULATION - Behavioral Simulation - Functional - sim_1 - tb_prime
prime.sv
tb prime.sv
Untitled 4*
C:/Users/manar/project_8/project_8.srcs/sim_1/new/tb prime.sv
`timescale 1ns/1ps
module tb_prime;
logic [2:0] A;
logic isPrime;
logic expected;
// Instantiate DUT
prime DUT (
.A (A),
.isPrime (isPrime)
);
// Reference model based on the truth table in the task
function automatic logic is_prime_ref(input logic [2:0] val);
case (val)
3'b000: is_prime_ref = 1'b0;
3'b001: is_prime_ref = 1'b0;
3'b010: is_prime_ref = 1'b1;
3'b011: is_prime_ref = 1'b1;
3'b100: is_prime_ref = 1'b0;
3'b101: is_prime_ref = 1'b1;
3'b110: is_prime_ref = 1'b0;
3'b111: is_prime_ref = 1'b1;
default: is_prime_ref = 1'bx;
endcase
endfunction
integer i;
integer errors = 0;
initial begin
$display("--");

Task #1.3

Using SystemVerilog, implement the following design and <u>(USE the design sources below) then write testbench to verify the comparator</u>

A, B: unsigned binary numbers
If A==B, then AeqB=1, AltB=0, AgtB=0
If A<B, then AeqB=0, AltB=1, AgtB=0
If A>B, then AeqB=0, AltB=0, AgtB=1



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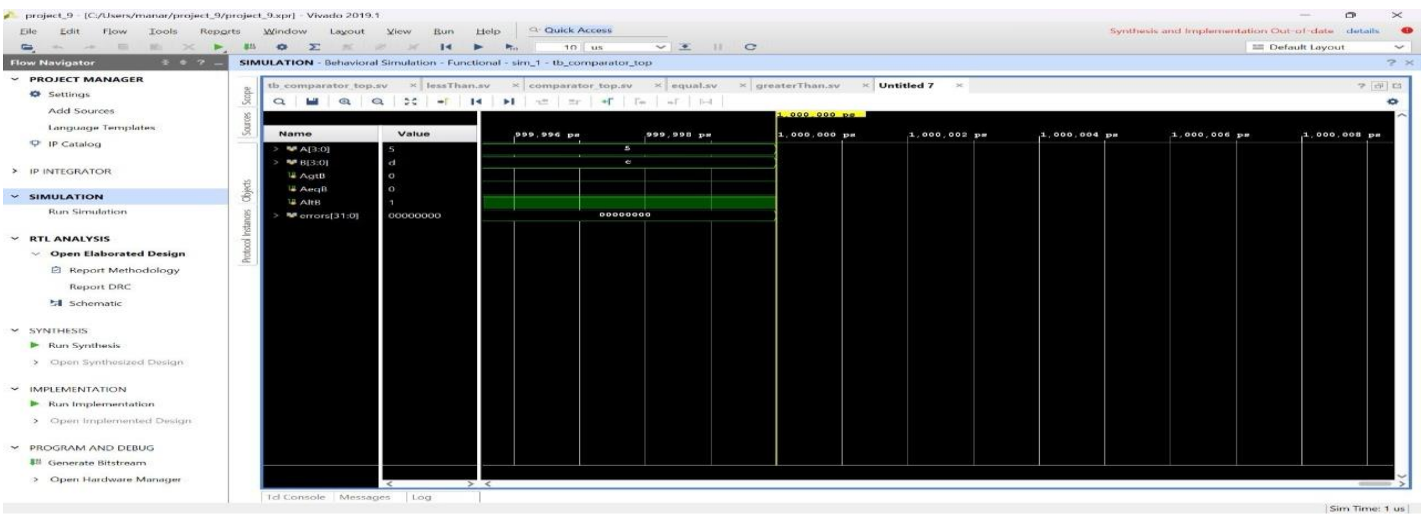
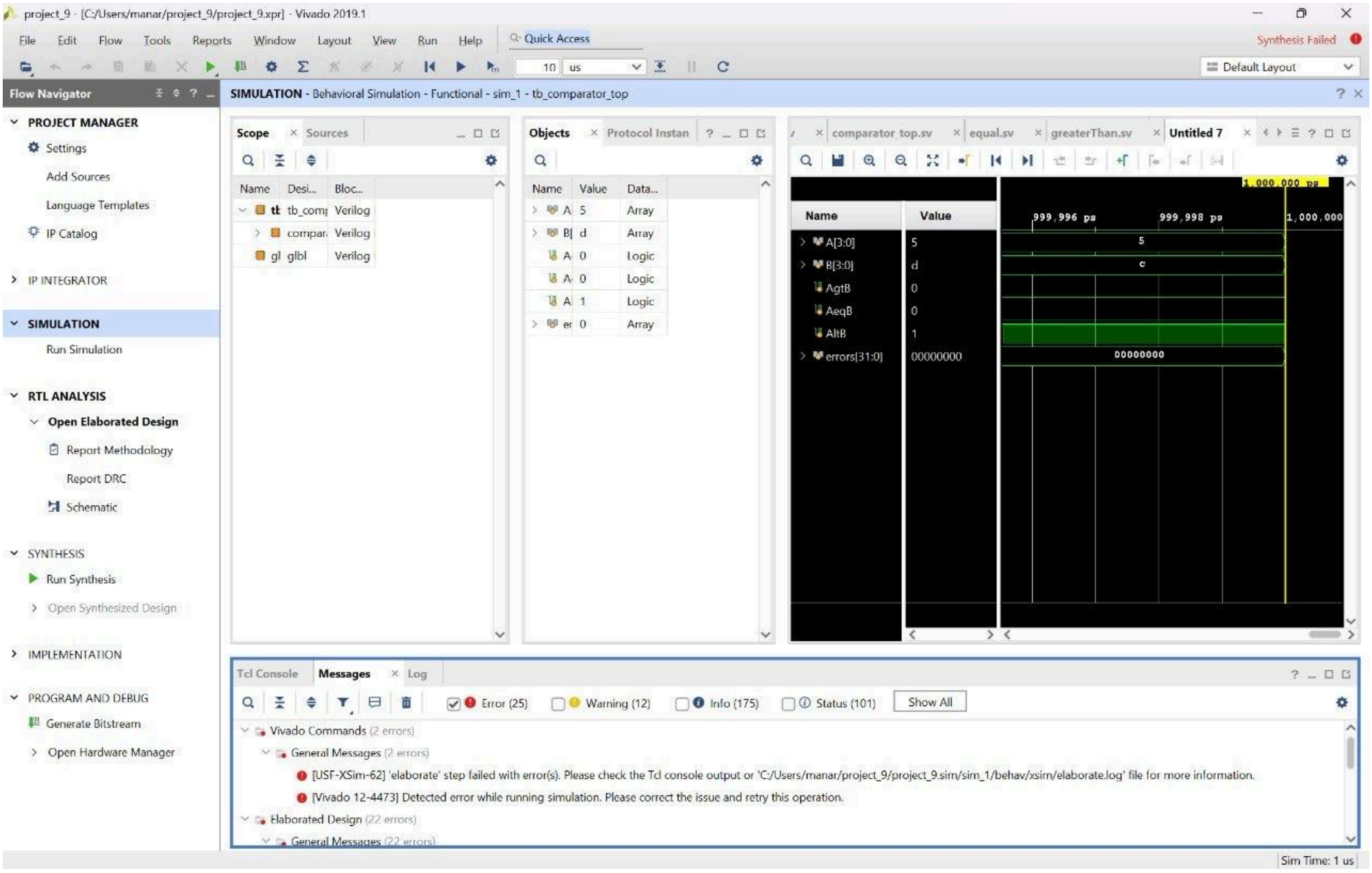
lessThan.sv:

```

module lessThan(AltB, a, b);

input [3:0] a,b;
output AltB;

assign AltB = (A < B)? 1 : 0;
endmodule

```




project_9 - [C:/Users/manar/project_9/project_9.xpr] - Vivado 2019.1
File Edit Flow Tools Reports Window Layout View Run Help
Quick Access
Synthesis Failed
Default Layout
Flow Navigator
SIMULATION - Behavioral Simulation - Functional - sim_1 - tb_comparator_top
PROJECT MANAGER
Settings
Add Sources
Language Templates
IP Catalog
IP INTEGRATOR
SIMULATION
Run Simulation
RTL ANALYSIS
Open Elaborated Design
Report Methodology
Report DRC
Schematic
SYNTHESIS
Run Synthesis
Open Synthesized Design
IMPLEMENTATION
PROGRAM AND DEBUG
Generate Bitstream
Open Hardware Manager
Scope
Sources
Objects
Protocol Instan
comparator_top.sv
equal.sv
greaterThan.sv
Untitled 7
Name
Value
A[3:0] 5
B[3:0] d
AgtB 0
AeqB 0
AltB 1
errors[31:0] 00000000
999,996 ps
999,998 ps
1,000,000 ps
Tcl Console
Messages
Log
Error (25)
Warning (12)
Info (175)
Status (101)
Show All
Vivado Commands (2 errors)
General Messages (2 errors)
[USF-XSim-62] 'elaborate' step failed with error(s). Please check the Tcl console output or 'C:/Users/manar/project_9/project_9.sim/sim_1/behav/xsim/elaborate.log' file for more information.
[Vivado 12-4473] Detected error while running simulation. Please correct the issue and retry this operation.
Elaborated Design (22 errors)
General Messages (22 errors)
Sim Time: 1 us